Learned Deferral Did Not Beat Confidence in a Same-Model Thinking Cascade

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Abstract

Hybrid reasoning models such as Qwen3 can answer directly or think first, so a cheap policy is to draft an answer directly and pay for a thinking solve only when the draft looks unreliable. We ask whether a learned deferral rule chooses those escalations better than the standard baseline, a threshold on the draft's token-level confidence, when the utility charges for both calls. On 50 GSM8K questions with eight independent draws per mode, Qwen3-1.7B's thinking solve fixes 30 of 400 direct drafts and breaks 10. Yet neither a predeclared learned rule, nor nine follow-up rules (stronger regularization, contrastive objectives, released contrastive-language-model heads, and token-level uncertainty), nor a simulated two-draft agreement rule improves on confidence for held-out questions. The predeclared learner has lower utility than confidence at every cost preference we tested; once refitting variation is included, its interval includes zero, so it fails to beat confidence without being reliably worse. Confidence itself captures only 17.6% of the hindsight oracle's gain. For Qwen3-0.6B, deferral beats random escalation at equal cost by 6.2 accuracy points, but most of its utility gain over accepting the draft comes from thinking itself; for Qwen3-1.7B, confidence deferral does not pay. The panels are small and were reused across analyses; we report every predeclared criterion, including those that failed.

1 Introduction

Hybrid reasoning models such as Qwen3 [1] offer a direct mode and a thinking mode in one set of weights. Thinking is usually more accurate but costs several times as many output tokens. A natural policy is to answer directly first and to think only when the direct draft looks unreliable. In the language of model cascades, the direct mode is the first stage, the thinking mode is the second, and a deferral rule decides when to call the second stage [2, 3]. Unlike most language-model cascades, both stages here are the same model.

Prior work on cascades gives a reason to expect a simple confidence rule to be hard to beat. Jitkrittum et al. [3] characterize the optimal deferral rule and show that confidence is highly competitive on many regular classification tasks. Learned post-hoc rules can significantly improve on it when the second model is a specialist, when labels are noisy, or when the distribution shifts, but may offer little when the second model's outputs are hard to predict. Gupta et al. [4] find that sequence-level uncertainty in generative cascades is length-biased and that learned deferral on token-level uncertainty quantiles helps. For deciding when to reason, Lewis-Lim et al. [5] find that training-free confidence measures can reduce redundant reasoning but are inconsistent: for some models, including Qwen3-8B, no measure consistently beats random gating, and GSM8K shows limited benefit.

We test whether learned post-hoc deferral improves on confidence in this same-model cascade, under a fixed contract. A small Qwen3 model first writes a direct draft. A rule then accepts the draft or requests a fresh thinking solve of the same question; the thinking call does not see the draft. Because the draft is already paid for, escalation charges both calls. We collect eight independent responses per mode for each of 50 GSM8K questions [6], fit rules on some questions, and evaluate them on others.

Our findings are as follows.

- **Observed opportunities did not become better deferral.** Thinking rescues 30 of 400 drafts (a wrong draft followed by a correct thinking solve), yet the predeclared learner fails four of its five success criteria on held-out questions, and nine follow-up rules and a two-draft agreement rule also fail to improve on confidence (Table 1).

- **The comparison is not specific to one cost.** Refitted at ten cost preferences, the predeclared learner has lower utility than confidence at every one, and most learned rules lie below random escalation at equal cost (Figure 1).
- **The gap is a missing gain, not a demonstrated loss.** For the predeclared learner, intervals that hold fitted actions fixed exclude zero, but intervals that include refitting variation do not (Table 2).
- **Whether deferral pays depends on the solver.** For Qwen3-0.6B, confidence deferral beats random escalation by 6.2 points at equal cost, and its gain over accepting the draft stays positive with refitting, though most of that gain comes from thinking itself. For Qwen3-1.7B it does not pay. Lower confidence is associated with higher gain only for the smaller model.

None of the ingredients is new: confidence-gated thinking [5], post-hoc deferral after an observed draft [3, 4], agreement-based routing [7, 8], and learned routers (Section 5) all precede this work. Our contribution is a controlled test of learned deferral in the case where both stages are the same small hybrid model, with repeated draws per question, held-out questions, a utility that charges the draft, intervals that include refitting variation, and every predeclared criterion reported. The answer is negative: confidence was not beaten, which fits the caveats in the cascade analysis about when post-hoc rules may not help, and the larger practical lever was whether thinking pays for the solver at all.

2 Setup

Decision and utility. Let D be a completed direct response and T a fresh thinking response, with scored correctness $Y_D, Y_T \in \{0, 1\}$ and charged output tokens C_D, C_T (including inserted closure tokens). Action $a = 1$ requests thinking and $a = 0$ accepts the draft. The utility and the gain from deferring are

$$U(a) = Y_D + a(Y_T - Y_D) - \lambda(C_D + aC_T)/1000, \qquad G = Y_T - Y_D - \lambda C_T/1000, \tag{1}$$

with $\lambda = 0.05$ fixed before the studies. Utility is measured in correct answers per question: a difference of $+0.01$ is one more correct answer per 100 questions, and $\lambda = 0.05$ prices 20,000 output tokens at one correct answer. Because U is linear in a, a rule that knew $\mathbb{E}[G \mid x, D]$ would defer exactly when it is positive, which parallels the optimal deferral rule of Jitkrittum et al. [3]. A confidence rule defers when the draft's mean token log probability, computed after the sampling transformations (temperature, top-k, top-p), falls below a threshold chosen on training questions only. It can approach the optimal rule only if lower confidence means higher expected gain, an analogue of their Lemma 4.1. A learned post-hoc rule instead estimates G from permitted features: the question and the observed draft, but never the thinking output, its length, or the reference answer.

Data and solvers. All questions are GSM8K training rows assigned to frozen roles; the official test split stays sealed. The main panel (50 questions, eight draws per mode) uses Qwen3-1.7B with 2,048 reasoning tokens and 512 answer tokens, sampled with the recommended settings. A separate two-solver panel (64 other questions) runs Qwen3-0.6B and Qwen3-1.7B with two draws per mode and solver. Correctness comes from a frozen numeric answer scorer (numeric-v7); rejected or truncated outputs count as wrong, and no human scoring audit was performed. Reference answers were opened only after each study's full set of responses was complete.

Evaluation. Each main-panel question has eight independently seeded responses per mode. Pairing the direct and thinking responses with the same draw index gives 400 draft–thinking pairs, which we call states. Five outer folds of ten questions keep every evaluation question out of fitting and tuning. A rule is fitted on draws 0–3 of the 40 training questions and evaluated on draws 4–7 of the 10 held-out questions, and vice versa; we report the average of the two directions. Four inner question folds choose settings. Intervals use 2,000 whole-question bootstrap resamples (90%) with fitted actions held fixed, unless stated otherwise.

Table 1: Main panel, Qwen3-1.7B, $\lambda = 0.05$, averaged over both draw directions (400 states). Tokens include the paid draft. Differences are against confidence, with 90% whole-question intervals holding fitted actions fixed. Lift is accuracy minus random escalation at the same token cost, in percentage points. †Derived from the reported accuracy and tokens. Rows without an interval were not bootstrapped against confidence.

Rule	Accuracy	Tokens	Utility vs. confidence [90%]	Lift
Always direct	89.50%	221.14	−0.012 [−0.034, +0.007]	0.00
Always think	94.50%	1,610.97	−0.032	0.00
Hindsight oracle (not deployable)	97.00%	342.02	+0.057	+7.07
Confidence	**92.75%**	**628.47**	reference	**+1.78**
Predeclared learner (combined Ridge)	89.00%	757.13	−0.044 [−0.064, −0.025]	−2.43
Its training-selected comparator	90.25%	683.06	−0.028	−0.91
Same Ridge, draft features only	89.75%	621.67	−0.030	−1.19
Same Ridge, question features only	88.75%	794.10	−0.048	−2.81
Regularized/compressed (CPU family)	90.75%	514.22	−0.014 [−0.026, −0.004]	+0.20
Compact neural objectives	89.25%	604.82	−0.034 [−0.053, −0.017]	−1.63
Released CLM, zero-shot	94.75%	1,550.53	−0.026 [−0.059, +0.017]	+0.47
Released CLM, calibrated	88.50%	443.71	−0.033 [−0.054, −0.015]	−1.80
Same encoder, PCA8/Ridge	89.00%	330.81	−0.023 [−0.044, −0.004]	−0.89
CLM/raw learned family	88.75%	406.34	−0.029 [−0.050, −0.011]	−1.42
Token-level learned error score	88.75%	570.33	−0.037 [−0.061, −0.013]	−2.01
Minimum-token threshold	91.00%	483.20	−0.010 [−0.027, +0.005]	+0.56
Token-level gain Ridge	90.00%	483.21	−0.020 [−0.035, −0.007]	−0.44
Two-draft agreement (simulated)	92.25%	597.69	−0.003 [−0.021, +0.012]	+1.40†

Rules. The predeclared learner (the combined learner in Figure 1) is Ridge regression of G on 387 question features (MiniLM embeddings [9] and three lexical counts) and four draft features (confidence, log length, parser validity, truncation); it defers when predicted gain is positive. Its comparator is the best of accept, escalate, and confidence, chosen within training. Later bounded studies, each with its own frozen grid and success criterion, tested stronger regularization and compression, compact contrastive and cross-entropy objectives, released contrastive-language-model (CLM) heads [10] with a matched-encoder control, token-level uncertainty quantiles adapted from Gupta et al. [4], and a two-draft agreement rule adapted from DART [8]. Two of these follow-up rules, zero-shot CLM and the minimum-token threshold, involve no learned weights beyond a threshold. Each follow-up rule had to beat confidence by at least +0.01 utility, with positive means in both directions and positive leave-one-question-out averages. Appendix A gives the details.

3 Results

3.1 No tested rule beats confidence

Table 1 lists the rules on the main panel at $\lambda = 0.05$. Direct accuracy is 89.5% and thinking accuracy 94.5%. Of the 400 paired states, 30 are rescues, 10 are harms, 348 have both responses correct, and 12 have both wrong.

The predeclared learner's utility difference against its training-selected comparator is −0.016 (90% interval $[-0.033, -0.001]$), and four of its five success criteria fail. Against confidence it is lower by 0.044. The same Ridge on the four draft features alone, which include confidence, is also lower (−0.030), so the 387 question features do not explain the loss by themselves. Every follow-up rule also has lower point utility than confidence, and every success criterion fails. The closest rules are two-draft agreement (−0.003), which escalates only 8.5% of states, and a threshold on the lowest token log probability (−0.010).

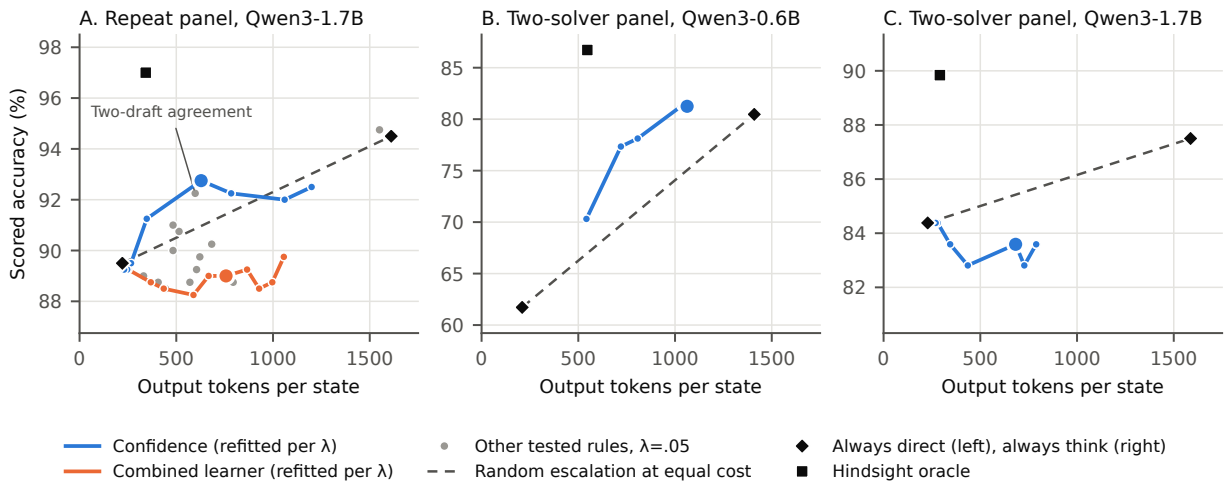


Figure 1: Accuracy and output cost of the tested deferral rules. In each panel, the dashed line is random escalation at equal cost between always accepting the draft and always thinking, and the square is the hindsight oracle. Lines trace rules refitted at ten cost preferences from $\lambda = 0$ to 0.3; large dots mark $\lambda = 0.05$. The combined learner is the predeclared learner. Panels use different axes. Values are post-hoc, on reused development panels, and carry no intervals.

Table 2: 90% intervals with fitted actions held fixed and with refitting included, for utility contrasts at $\lambda = 0.05$. Copied questions in the refit bootstrap reduce the number of distinct training questions, so those intervals are approximate. The last column gives the 5th–95th percentiles over 200 fold repartitions.

Contrast	Point	Fixed actions	With refitting	Fold repartitions
Learner − training-selected comparator	−0.016	[−0.033, −0.001]	[−0.042, +0.021]	[−0.036, −0.005]
Learner − confidence	−0.044	[−0.064, −0.025]	[−0.058, +0.008]	[−0.051, −0.029]
Confidence − accept (1.7B, main panel)	+0.012	[−0.007, +0.034]	[−0.027, +0.028]	[+0.003, +0.015]
Confidence − accept (0.6B, two-solver)	+0.153	[+0.086, +0.220]	[+0.068, +0.220]	[+0.131, +0.168]
Confidence − accept (1.7B, two-solver)	−0.031	[−0.056, −0.006]	[−0.051, +0.010]	[−0.037, −0.016]

3.2 The result holds across costs

Figure 1A places each rule on accuracy and cost. Random escalation at a fixed rate traces the dashed line between always direct and always think. Confidence lies 1.78 points above that line at $\lambda = 0.05$, while the predeclared learner lies 2.43 points below it. We refitted the original rules at ten values of λ from 0 to 0.3. The predeclared learner has lower utility than confidence at all ten (by 0.006 to 0.054) and lower than its training-selected comparator at all ten. Among the follow-up rules, only the minimum-token threshold, zero-shot CLM, and the CPU family lie above the random line, each by less than 0.6 points. The single cost preference of the original studies therefore does not explain the result.

3.3 With refitting, the gap is a missing gain

Fixed-action intervals treat each fitted rule as given. To include the variation from fitting, we resampled whole questions 2,000 times and refitted the rules on each resample. Each copy of a question kept its original fold, and inner folds were formed over distinct questions, so no question appeared in both training and evaluation. Separately, 200 random repartitions of the questions into folds isolate the effect of fold assignment.

Refitting widens the intervals by 1.1 to 1.9 times (Table 2). The intervals for the learner's gap to both comparators now include zero, as does the 1.7B confidence loss on the two-solver panel. The fold assignment does not drive the signs: every repartition range stays on the side of its point estimate. We conclude that the predeclared learner did not deliver its predeclared gain of at least +0.01; the data do not show that it

is worse than confidence. Only this learner and confidence were refitted; the follow-up rules' fixed-action intervals in Table 1 would also widen if refitting were included.

3.4 Deferral pays for the smaller solver only

On the two-solver panel (Figure 1B–C), Qwen3-0.6B's direct accuracy is 61.7% and its thinking accuracy 80.5%. Confidence deferral reaches 81.25% at 1,063 tokens per state, 6.2 points above random escalation at equal cost, and gains +0.153 utility over accepting the draft. That gain stays clear of zero with refitting and is positive at every λ up to 0.2, but most of it is the value of thinking: always thinking after the draft already gains +0.128, and confidence leads always thinking by only +0.025, a difference without an interval. For Qwen3-1.7B, direct accuracy is 84.4% and thinking 87.5%. Confidence deferral is 1.8 points below random escalation at equal cost and has lower utility than accepting the draft at every λ tested. On the main panel, its gain over accepting the 1.7B draft (+0.012) is not distinguishable from zero. The two-draft agreement rule also fails on this panel: it trails confidence by 0.028 for Qwen3-0.6B, and for Qwen3-1.7B it trails accepting the draft by 0.023.

4 Why confidence was hard to beat

Few opportunities, many costly mistakes. At $\lambda = 0.05$ the average thinking call costs about 0.07 utility, so more than about one escalation in fourteen must be a rescue before harms are counted. The hindsight oracle escalates exactly the 30 rescues (7.5% of states) and reaches 97.0% at 342 tokens. Confidence escalates 100 states: 17 rescues, 4 harms, 4 pairs in which both responses are wrong, and 75 pairs in which both were already correct. It therefore captures 17.6% of the oracle's gain. The predeclared learner escalates 162 states but rescues only 4.

Where the rescues sit. A post-hoc audit grouped states into five confidence bins. For Qwen3-0.6B, mean gain falls at every step from the lowest-confidence bin to the highest, and the two lowest bins hold 23 of 32 rescues: confidence is the right ranking. For Qwen3-1.7B on the main panel, only the lowest-confidence fifth has positive mean gain (+0.081, interval $[-0.016, +0.176]$), and 13 of the 30 rescues sit in higher bins, outnumbered by states whose only gain is the thinking cost. The highest-confidence bin holds 8 of the 12 pairs in which both responses are wrong; these drafts are confidently wrong and thinking does not fix them. On the 1.7B two-solver panel, no confidence threshold improves on accepting the draft even in-sample. A learned rule would need features that find the remaining rescues among many already-correct drafts, from 40 training questions per fold and 391 features. None of the tested rules did.

What cascade analysis does and does not explain. Confidence traces the same deferral curve as the optimal rule when it orders inputs the same way as the gain from deferring (Lemma 4.1 of Jitkrittum et al. [3]). The bins above suggest that an analogue of this holds roughly for Qwen3-0.6B but not for Qwen3-1.7B, so we have no reason to think confidence is near-optimal on the larger solver, although the hindsight oracle overstates the usable headroom because it sees the realized thinking response. The three settings in which they show post-hoc rules helping are at most partly present. The second stage is the same model and more accurate on average (94.5% against 89.5%), but its errors are not uniform: on the 1.7B two-solver panel, 12 of the 18 thinking responses that hit the reasoning cap are wrong, against 4 of the other 110. A rule cannot see a cap hit before paying for thinking, and we did not test directly whether question or draft features predict it. Training and evaluation questions come from the same pool, so there is no designed shift. We injected no label noise. In their label-noise case both models fail on low-confidence inputs, so confidence wastes deferrals on them. The closest analogue here, pairs in which both responses fail, differs by panel: on the main panel 8 of 12 sit in the highest-confidence bin, where a confidence rule does not escalate, but on the 1.7B two-solver panel 10 of 13 sit in the two lowest bins, where it does. They also warn that post-hoc rules may devolve to plain confidence deferral when the second model's outputs are not strongly predictable, and their excess-risk bound has an estimation term that shrinks with the size of the fitting sample. Both caveats fit this setting: the predeclared test that question-level gain is reproducible did not pass, and each fold trains on 40 questions. This reading is consistent with the failures but was not tested directly.

Token-level features added no usable signal. Gupta et al. [4] find that summing token log probabilities over-defers long outputs and that averaging over-corrects toward short ones. Our confidence is the average, and it is nearly uncorrelated with answer length (Spearman -0.027). Low token quantiles rank errors slightly better in-sample (positive-gain AUROC 0.761 against 0.705) without improving held-out decisions. On accuracy tasks their rule is a five-layer MLP trained to predict that the large model is right and the small one wrong; with 40 training questions per fold we fitted logistic and Ridge models instead, so ours is not a test of their model class or target.

Agreement is priced out. Two-draft agreement nearly matches confidence while escalating a third as often. If the second draft were free, it would lead confidence by 0.008; at $\lambda = 0.05$ its cost, about 0.011 utility per state, removes the advantage. Settings where a second draft is cheap relative to thinking were not tested.

5 Related work

Cascades and deferral. FrugalGPT learns when to continue to a stronger model [2]; Yue et al. [7] defer on the consistency of sampled answers; AutoMix uses self-verification and a POMDP router [11]; Fanconi and van der Schaar [12] place an accept-or-regenerate decision before human escalation. Jitkrittum et al. [3] analyze when confidence-based deferral suffices, and Gupta et al. [4] study token-level uncertainty for generative cascades. Our primary learner is close to the Diff-01 post-hoc estimator of Jitkrittum et al. [3], a squared-error regression on the difference in correctness, with token cost added to the target.

Adaptive thinking. Lewis-Lim et al. [5] gate chain of thought on direct-answer confidence, with Qwen3-8B/32B and two GPT models on seven benchmarks including GSM8K. Benefits depend on the dataset and model, which matches our solver dependence, and their perplexity measure is also a transform of a mean token log probability. Their direct mode outputs only a final answer, whereas our direct drafts average 221 tokens, so the settings are not directly comparable. We add learned rules and repeated draws per question, and we charge both calls on every escalation; their released analysis code counts only the reasoning response. DART routes after observing two direct drafts [8]; our agreement rule simulates its first stage without budget probing and is not a faithful comparison. Self-Route [13], Thinkless [14], and BEST-Route [15] choose modes or sample counts before generation.

Routing evaluation and representations. Recent audits caution that routing headroom need not be usable value [16, 17], and that predicting failure differs from predicting which intervention pays [18]. RouterDC [19], cost-aware contrastive routing [20], and SEMIROUTER [21] precede our use of repeated labels and contrastive representations.

6 Limitations

The study uses one dataset on which the larger solver is near ceiling, and two small solvers from one family. The predeclared learner and all follow-up rules reuse the same 50 questions, and later analyses, including the cost curves, refit intervals, confidence bins, token-level rules, and agreement simulation, are post-hoc on those panels. The intervals that include refitting cover five contrasts and are approximate; the other intervals hold fitted actions fixed. Correctness is scorer-based, with no human audit. The token-level and agreement rules are adaptations, not reproductions, and we compare against no external system. Utility counts output tokens only, not input processing, encoder work, or latency. Two later interventions, a parser guard tested on 228 fresh questions and a larger reasoning allowance on 11 selected questions, each produced at most one rescue (Appendix B); they do not change these conclusions.

7 Conclusion

In a same-model cascade from a direct draft to a fresh thinking solve, observed rescues did not become better learned deferral. Confidence had the highest point utility among the deferral rules we tested on the main

panel, yet it captured 17.6% of the hindsight gain, and it paid only for the smaller solver. The settings in which post-hoc rules are known to help were at most partly present, each fold trained on 40 questions, and the predeclared test that question-level gain is reproducible did not pass. Tasks with more rescue opportunities, cheaper second stages, or real distribution shift are where a learned rule would have room to win, and a larger prospective study would be needed to show it.

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Code and data availability. Code, frozen protocols, and a cache of scored outcomes, draft features, and recorded decisions will be released after review. A check script recomputes every value in Tables 1 and 2, every point in Figure 1, and the values quoted in the text from that cache alone (195 checks, all passing). GSM8K is released under the MIT license and the Qwen3 models under Apache-2.0; no human subjects or personal data were involved.

References

[1] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jing Zhou, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghao Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 technical report, 2025. URL https://arxiv.org/abs/2505.09388v1.

[2] Lingjiao Chen, Matei Zaharia, and James Zou. FrugalGPT: How to use large language models while reducing cost and improving performance, 2023. URL https://arxiv.org/abs/2305.05176.

[3] Wittawat Jitkrittum, Neha Gupta, Aditya Krishna Menon, Harikrishna Narasimhan, Ankit Singh Rawat, and Sanjiv Kumar. When does confidence-based cascade deferral suffice? In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://arxiv.org/abs/2307.02764.

[4] Neha Gupta, Harikrishna Narasimhan, Wittawat Jitkrittum, Ankit Singh Rawat, Aditya Krishna Menon, and Sanjiv Kumar. Language model cascades: Token-level uncertainty and beyond. In *International Conference on Learning Representations*, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/file/11f5520daf9132775e8604e89f53925a-Paper-Conference.pdf.

[5] Samuel Lewis-Lim, Xingwei Tan, Zhixue Zhao, and Nikolaos Aletras. Can confidence estimates decide when chain-of-thought is necessary for LLMs?, 2026. URL https://arxiv.org/abs/2510.21007.

[6] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems, 2021. URL https://arxiv.org/abs/2110.14168v2.

[7] Murong Yue, Jie Zhao, Min Zhang, Liang Du, and Ziyu Yao. Large language model cascades with mixture of thought representations for cost-efficient reasoning. In *International Conference on Learning Representations*, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/5de11e930c1bbfda5d4fc9d2b0924032-Abstract-Conference.html.

[8] Jungseob Lee, Seongtae Hong, Seungjun Lee, Jaehyung Seo, Junyoung Son, Sugyeong Eo, Chanjun Park, Hyeongju Park, Hyeonseok Moon, and Heuiseok Lim. DART: Draft-agreement routing for training-free adaptive thinking budgets in hybrid reasoning models, 2026. URL https://arxiv.org/abs/2606.23181v4.

[9] Sentence Transformers. all-MiniLM-L6-v2: Official model card. Hugging Face model card. URL https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2.

[10] Jacky Kwok, Hangoo Kang, Tarun Suresh, Jon Saad-Falcon, Marco Pavone, Christopher Ré, and Azalia Mirhoseini. Contrastive language models: A system one model for fast and generalizable decision-making. Author blog and released implementation, 2026. URL https://contrastive-lm.notion.site.

[11] Pranjal Aggarwal, Aman Madaan, Ankit Anand, Srividya Pranavi Potharaju, Swaroop Mishra, Pei Zhou, Aditya Gupta, Dheeraj Rajagopal, Karthik Kappaganthu, Yiming Yang, Shyam Upadhyay, Manaal Faruqui, and Mausam. AutoMix: Automatically mixing language models. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/ecda225cb187b40ea8edc1f46b03ffda-Abstract.html.

[12] Claudio Fanconi and Mihaela van der Schaar. Cascaded language models for cost-effective human–AI decision-making. In *Advances in Neural Information Processing Systems*, 2025. URL https://arxiv.org/abs/2506.11887v3.

[13] Yang He, Xiao Ding, Bibo Cai, Yufei Zhang, Kai Xiong, Zhouhao Sun, Bing Qin, and Ting Liu. Self-Route: Automatic mode switching via capability estimation for efficient reasoning, 2025. URL https://arxiv.org/abs/2505.20664.

[14] Gongfan Fang, Xinyin Ma, and Xinchao Wang. Thinkless: LLM learns when to think, 2025. URL https://arxiv.org/abs/2505.13379v2.

[15] Dujian Ding, Ankur Mallick, Shaokun Zhang, Chi Wang, Daniel Madrigal, Mirian Del Carmen Hipolito Garcia, Menglin Xia, Laks V. S. Lakshmanan, Qingyun Wu, and Victor Rühle. BEST-Route: Adaptive LLM routing with test-time optimal compute, 2025. URL https://arxiv.org/abs/2506.22716v1.

[16] Teng-Ruei Chen. What does a routing oracle measure under stochastic decoding? coupling, scorer choice, and single-commit ceilings, 2026. URL https://arxiv.org/abs/2607.03436v3.

[17] Saloni Garg and Amit Sagtani. Unsolvability ceiling in multi-LLM routing: An empirical study of evaluation artifacts, 2026. URL https://arxiv.org/abs/2605.07395v1.

[18] Chih-Hsuan Yang, Jingyan Jiang, Cheng-Hau Yang, Vikram Vasudevan, Huihuo Zheng, Venkatram Vishwanath, and Rajeev Thakur. LLMs can predict failure risk, but struggle to predict which collaboration protocol pays off: Cost-aware protocol routing across reasoning tasks, 2026. URL https://arxiv.org/abs/2608.14927v1.

[19] Shuhao Chen, Weisen Jiang, Baijiong Lin, James T. Kwok, and Yu Zhang. RouterDC: Query-based router by dual contrastive learning for assembling large language models. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-2120. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/7a641b8ec86162fc875fb9f6456a542f-Abstract-Conference.html.

[20] Reza Shirkavand, Shangqian Gao, Peiran Yu, and Heng Huang. Cost-aware contrastive routing for LLMs, 2025. URL https://arxiv.org/abs/2508.12491v3.

[21] Zijie Wang, Xinyu Yan, Che Wang, Zeng Zihao, Lei Xiao, and Wei Yang Bryan Lim. SEMIROUTER: Sparse-data enhanced routing for adaptive multi-LLM system. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 4910–4921, Rabat, Morocco, March 2026. Association for Computational Linguistics. doi: 10.18653/v1/2026.eacl-long.228. URL https://aclanthology.org/2026.eacl-long.228/.

A Rules and settings

Models and generation. Qwen3-1.7B (revision 70d244cc) and Qwen3-0.6B (revision c1899de2), local float16 inference with batch size one. Direct sampling: temperature 0.7, top-p 0.8, top-k 20. Thinking sampling: temperature 0.6, top-p 0.95, top-k 20. Reasoning allowance 2,048 tokens, answer allowance 512 tokens; forced closure, when needed, is charged. The prompt asks for the numerical answer in `\boxed{}`.

Predeclared learner. Combined Ridge on 391 features, penalty chosen from $\{1, 10, 100\}$ by inner question folds, with training-only standardization. The comparator is chosen within training from accept, escalate, and a confidence threshold. Training conditions use 1, 2, or 4 draws per training question; the four-draw average across directions is primary, and an eight-draw condition is secondary. The five criteria were a positive reproducibility test of question-level gain, positive gains in both four-draw directions, an average gain of at least $+0.01$ with a positive lower bound, no reversal in the eight-draw condition, and at least ten informative questions; only the last passed.

Follow-up rules. *CPU family:* Ridge penalties $\{10^2, 10^3, 10^4\}$, PCA compression to eight question components, and conservative confidence corrections, 1,035 Ridge fits in total. *Compact neural:* two representations crossed with four objectives (gain regression, regret-weighted cross-entropy, bidirectional contrastive, binary cross-entropy) on a 120-parameter head, one optimization seed. *CLM:* released heads on 4-bit Qwen3-8B last-token states, zero-shot and with scalar calibration, plus a PCA8/Ridge control on the same embeddings. *Token-level:* nine quantiles of teacher-forced token log probabilities, their mean, and log length; a logistic error score (primary), a minimum-token threshold, and gain Ridge. *Agreement:* each evaluation draft is paired with every other draw in its half; it is accepted if both parse to the same number and escalated otherwise; both drafts are charged. All follow-up rules select settings on training questions and draws only.

B Follow-up interventions

Parser guard on fresh questions. On the 0.6B development panel, escalating drafts that the scorer rejects added three rescues. A guard frozen before a fresh 228-question test changed one action; that thinking response was also rejected, giving a utility difference of -0.0003. The design identifies only the policy difference, and one event cannot settle the question.

Larger reasoning allowance. On 11 development questions selected for forced reasoning closure, raising the allowance from 2,048 to 4,096 tokens produced one rescue and no harms ($+4.55$ accuracy points) but cost 1,662 more thinking tokens, for a utility change of -0.038 (95% interval $[-0.095, +0.062]$).

C Reproducibility

Every study froze its protocol, grid, endpoints, and criteria before execution and recorded source, input, and output hashes. Generation charged 644,519 tokens for the main panel, 384,552 for the two-solver panel, 43,448 for the fresh guard test, and 127,171 for the allowance study, on local hardware. Earlier development studies used an older scorer and different questions; they are summarized, with all criteria and records, in an accompanying technical report that will be made available after review.